

THE STRUCTURE OF MATHEMATICAL REALITY

Phase IV — Chapter 13

Measurement: How the World Gets Numbers

Assignment

Self-Study Curriculum

Take one real dataset from your work. Produce a one-page “measurement audit”:

1. List each variable and assign it a scale type, naming the *empirical* operation that justifies the assignment (not just its appearance in the spreadsheet).
2. For each variable, list the operations your analysis actually performs on it (means, differences, ratios, correlations, distances).
3. Flag every place where an operation requires more structure than the scale carries. For each flag, decide: is it a genuine error, or a defensible interval-treatment of ordinal data? Justify.
4. Pick the most consequential flag and show, concretely, how an admissible transformation of the scale would change (or fail to change) your conclusion.
5. State the one measurement decision in this dataset that, if wrong, would most damage every downstream model.